

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and implemented **Python**-based ML models for user targeting and personalization, directly contributing to growth initiatives by improving engagement metrics by **15%**.
- Designed and built scalable **ML pipelines** on **AWS** using **Python** for data processing, model training, and deployment, handling approximately **4,000 events per second** during peak loads.
- Collaborated with **cross-functional teams**, including product managers and data scientists, to identify and scope potential ML solutions for user acquisition and retention strategies.
- Conducted **A/B tests** to evaluate the performance of newly developed ML models, optimizing their impact on key growth metrics and reporting findings with **clear statistical significance**.
- Analyzed large datasets using **SQL** and **Spark** to extract actionable insights that informed decision-making for user acquisition and retention strategies, leading to a **10% improvement** in conversion rates.
- Contributed to the development of internal **ML infrastructure**, ensuring it could support rapid experimentation and deployment of new models.

SKILLS

ML Engineering – Machine Learning, Model Development, Model Training, Model Deployment, Recommendation Systems, User Modeling, Personalization Algorithms, ML Pipelines, ML Infrastructure, MLOps, PyTorch, scikit-learn, Python

Data Analysis & Processing – SQL, Spark, Hadoop, Pandas, NumPy, Data Warehousing, ETL, Data Visualization, Statistical Analysis, A/B Testing, Time-Series Analysis

Programming & Scripting – Python, Java, Spring Boot, REST APIs, R

Cloud & DevOps – AWS, EC2, S3, Lambda, CloudWatch, Docker, Kubernetes, Jenkins, Git

Tools & Platforms – Jira, Confluence, Power BI, Tableau, Linux, Splunk, Dynatrace, FastAPI

PROJECTS

Growth Platform ML Model for User Engagement

- Developed and trained a **gradient-boosted classifier** using **Python** and **scikit-learn** on a dataset of approximately **350,000 user interaction records** to predict engagement likelihood.
- Engineered a **feature pipeline** in **Python** and **SQL** to extract user behavioural signals, feeding into the model training process and reducing feature engineering time by **30%**.
- Deployed the trained model as a **REST API** using **FastAPI** on **AWS Lambda**, achieving a p95 latency of **150ms** for real-time scoring.

Personalization Engine Enhancement for User Retention

- Designed and implemented a **recommendation system** component using collaborative filtering techniques in **Python**, processing over **1.5 million user-item interactions**.
- Collaborated with a product stakeholder to define key performance indicators and integrated **A/B testing** frameworks to measure the impact of the enhanced personalization on user retention rates.
- Built interactive **Power BI dashboards** to visualize model performance, user segment behaviour, and the impact of personalization on retention metrics for business review.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)